

# ALAKAZAM

## The in-battle coach — a plain brief

The final, hardest piece of ABRA, a system that learns competitive Pokémon from thousands of real games.

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### The problem

A competitive Pokémon battle is a **hidden-information, same-time game**: you can't see the opponent's full setup, and both players choose each turn at the same moment and reveal together. That mix — hidden cards plus simultaneous choices — is what makes poker hard, wrapped around a board that changes like chess. So the winning play often isn't one fixed move; it's a smart, slightly unpredictable mix.

### What ALAKAZAM will be

A coach that, at any point in a battle, tells you the best move to make and how likely you are to win — the equivalent of a chess engine's "best move + evaluation," but for this hidden, same-time game.

### How it works

- **A read on what's hidden** — a running best-guess of the opponent's setup, sharpened as the game reveals clues.
- **Looking ahead** — it imagines the next few turns (not the whole game, which is far too big) and scores where each line leads.
- **A fast judge of positions** — a small learned model that estimates "how likely am I to win from here," so it need not play every line to the end.
- **Staying human-like** — kept close to how real players play, so it's strong without drifting into weird, easily-punished habits.

### Compute needs

Using it is light — it runs in a web browser on a normal laptop, no special hardware. Building the strongest version is the heavy part: it learns from a huge pile of past matches and from playing millions of games against itself, which needs a powerful rented cloud computer (a GPU) for a while.

### How long

A basic but genuinely useful version: about **one to two weeks**. A version that could beat top human players: **a few months** — and most of that time is spent letting it play itself to learn, not the training run itself.

### The honest limit

Some things simply can't be predicted — like who wins from the starting line-ups alone, which is close to a coin flip. So ALAKAZAM is judged on making **good decisions in the moment**, not on calling the winner in advance.

ABRA · competitive-Pokémon decision engine · ALAKAZAM is the in-battle capstone (in development).